

Fourier Serileri
(Fourier Series)
Hafta7

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Textbook: Linear Systems and Signals, B.P. Lathi

This slides are modified from Linear Systems and Signals (EE313) course at the University of Texas at Austin, with Courtesy of Prof. Brian L. Evans

Spektrum Temsili

(Spectrum Representation)

- Sinusoidal waveforms**

$$x(t) = A \cos(2\pi f_0 t + \phi_0) = \operatorname{Re} \left\{ A e^{j(2\pi f_0 t + \phi_0)} \right\} = \operatorname{Re} \left\{ A e^{j\phi_0} e^{j2\pi f_0 t} \right\}$$

- Spectrum* is collection of amplitude, phase and frequency info to express more complicated signals**

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos(2\pi f_k t + \phi_k) = X_0 + \operatorname{Re} \left\{ \sum_{k=1}^N X_k e^{j2\pi f_k t} \right\}$$

where A_0 is real-valued and $X_0 = A_0$ and $X_k = A_k e^{j\phi_k}$

- Example (slide 1-16)**

$$\begin{aligned} y(t) &= x^2(t) = \cos^2(2\pi 440 t) \\ &= \frac{1}{2} (1 + \cos(2(2\pi 440 t))) \\ &= \frac{1}{2} + \frac{1}{2} \cos(2\pi 880 t) \end{aligned}$$

A_0			
A_1		A_2	
f_1	440 Hz	f_2	880 Hz
ϕ_1		ϕ_2	

İki-Taraflı Spektrum

(Two-Sided Spectrum)

- Expanding cosine terms with inverse Euler formula

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos(2\pi f_k t + \phi_k) = X_0 + \sum_{k=1}^N \left\{ \frac{X_k}{2} e^{j2\pi f_k t} + \frac{X_k^*}{2} e^{-j2\pi f_k t} \right\}$$

where $X_k = A_k e^{j\phi_k}$ and $X_k^* = A_k e^{-j\phi_k}$

- Two-sided spectrum

$2N+1$ frequencies and $2N+1$ complex amplitudes

$\{ (0, X_0), (f_1, \frac{1}{2} X_1), (-f_1, \frac{1}{2} X_1^*), \dots, (f_k, \frac{1}{2} X_k), (-f_k, \frac{1}{2} X_k^*), \dots \}$

Frequency-domain representation of signal $x(t)$

Two-Sided Spectrum Example

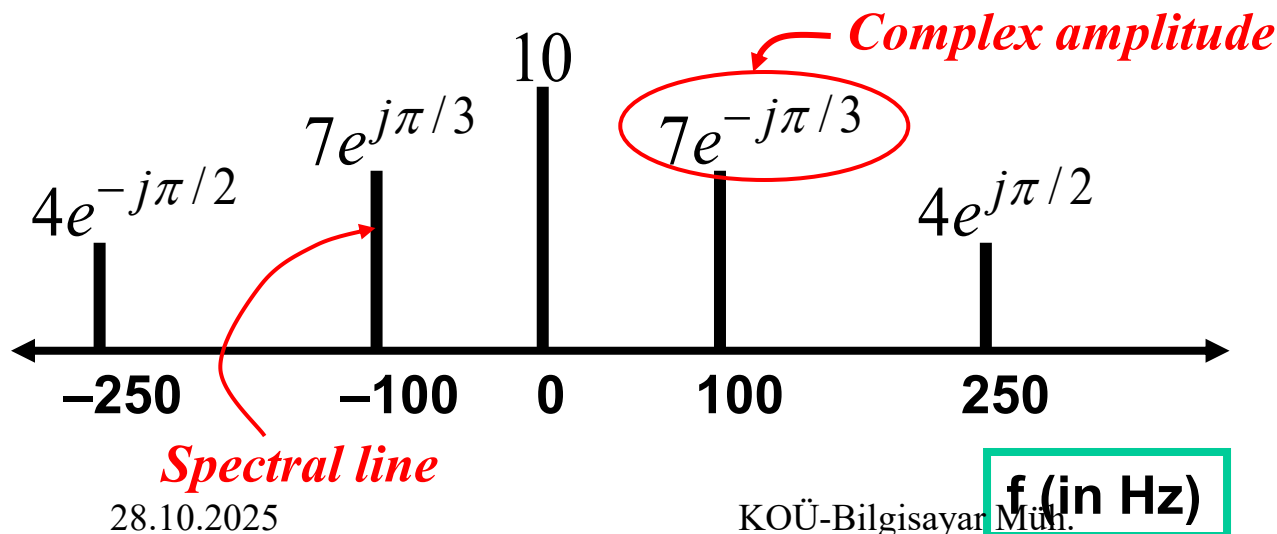
- $x(t) = 10 + 14 \cos(200\pi t - \pi/3) + 8 \cos(500\pi t + \pi/2)$

Expanding cosine terms using (inverse) Euler formula

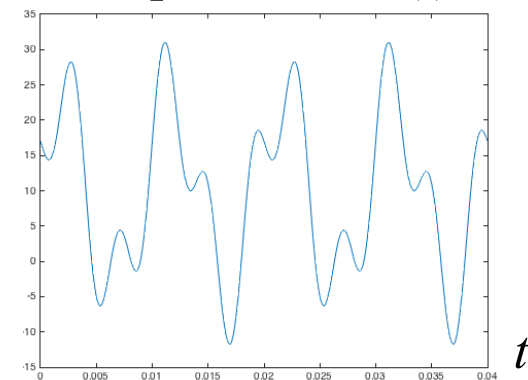
$$x(t) = 10 + 7e^{-j\pi/3} e^{j2\pi(100)t} + 7e^{j\pi/3} e^{-j2\pi(100)t} + 4e^{j\pi/2} e^{j2\pi(250)t} + 4e^{-j\pi/2} e^{-j2\pi(250)t}$$

Constant term has zero frequency: $10 e^{j0t} = 10$

- Spectrum plot**



Two periods of $x(t)$



Örnek: Ritim Notları

(Example: Beat Notes)

- Occurs when multiplying two sinusoidal signals

Audio: singing when holding a note, some musical instruments

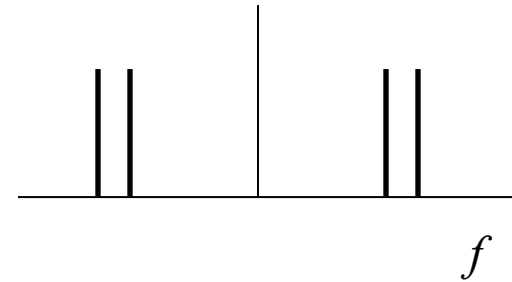
Communications: amplitude modulation (AM radio, Wi-Fi)

- Example:** $x(t) = \cos(2\pi 10t) \sin(2\pi 1000t)$

Rewrite $x(t)$ as sum of complex sinusoids

$$\begin{aligned}
 x(t) &= \frac{e^{j2\pi(10)t} + e^{-j2\pi(10)t}}{2} \frac{e^{j2\pi(1000)t} - e^{-j2\pi(1000)t}}{2j} \\
 &= \frac{1}{4j} e^{j2\pi(1010)t} - \frac{1}{4j} e^{-j2\pi(990)t} + \frac{1}{4j} e^{j2\pi(990)t} - \frac{1}{4j} e^{-j2\pi(1010)t} \\
 &= \frac{1}{2} \frac{1}{2j} e^{j2\pi(1010)t} - \frac{1}{2j} e^{-j2\pi(1010)t} + \frac{1}{2} \frac{1}{2j} e^{j2\pi(990)t} - \frac{1}{2j} e^{-j2\pi(990)t} \\
 &= \frac{1}{2} \sin(2\pi 1010t) + \frac{1}{2} \sin(2\pi 990t) = \frac{1}{2} \cos\left(2\pi 1010t - \frac{\pi}{2}\right) + \frac{1}{2} \cos\left(2\pi 990t - \frac{\pi}{2}\right)
 \end{aligned}$$

Spectrum

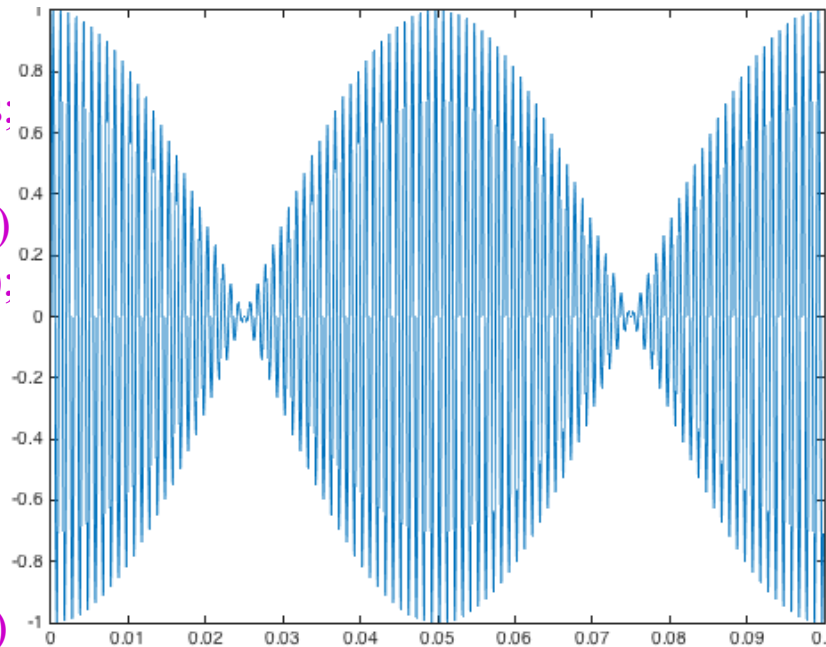


Example: Beat Notes (Continued)

- Example:** $x(t) = \cos(2\pi 10t) \sin(2\pi 1000t)$

Equal to:
$$x(t) = \frac{1}{2} \cos\left[2\pi 1010t - \frac{\pi}{2}\right] + \frac{1}{2} \cos\left[2\pi 990t - \frac{\pi}{2}\right]$$

```
f0 = 10; f1 = 1000;
fs = 8000; Ts = 1/fs;
t = 0 : Ts : 3;
x0 = cos(2*pi*f0*t)
x1 = sin(2*pi*f1*t);
x = x0 .* x1;
sound(x, fs);
% plot one period
T0 = 1/f0;
n = T0 / Ts;
plot( t(1:n), x(1:n) )
```



```
f2 = 1010; f3 = 990;
fs = 8000; Ts = 1/fs;
t = 0 : Ts : 3;
x2 = cos(2*pi*f2*t - pi/2);
x3 = cos(2*pi*f3*t - pi/2);
x = 0.5*x2 + 0.5*x3;
sound(x, fs);
% plot one period
f0 = gcd(f2, f3); T0 = 1/f0;
n = T0 / Ts;
plot( t(1:n), x(1:n) );
```

Periodic Signals

- A signal is periodic with period T_0 if $x(t + T_0) = x(t)$ for all t

Also periodic with periods $2T_0$, $3T_0$, etc., and $-T_0$, $-2T_0$...

Smallest positive period T_0 is called the *fundamental period*

Fundamental frequency f_0 is computed as $1 / T_0$

- A periodic signal $x(t)$

Unchanged when time-shifted by one period

May be generated by periodically extending one period

Area under $f(t)$ over any interval of duration equal to the period is same; e.g., integrating from 0 to T_0 would give the same value as integrating from $-T_0/2$ to $T_0/2$

Periodic Signals

$$\sin(2\pi f_0 t) = \sin(2\pi f_0 t + 2\pi) = \sin\left(2\pi f_0 \left(t + \frac{2\pi}{2\pi f_0}\right)\right) = \sin\left(2\pi f_0 \left(t + \frac{1}{f_0}\right)\right)$$

- **Synthesize periodic signals**

Add two or more cosine waves with harmonic frequencies

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos(2\pi f_k t + \phi_k) \text{ where the } k\text{th harmonic is } f_k = k f_0$$

- **Finding fundamental frequency**

Largest f_0 such that $f_k = k f_0$, i.e. $f_0 = \gcd\{f_k\}$

Consider notes A 440 Hz, E 660 Hz and F # 740 Hz. $f_0 = \underline{\hspace{1cm}}$

Notation and Speech Example

- **Alternate periodic signal synthesis formula**

Expand cosines into sum of two exponentials (Euler's formula)

Explicitly represents positive and negative frequencies:

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos(2\pi k f_0 t + \phi_k) = \sum_{k=-N}^N a_k e^{j2\pi k f_0 t} = a_0 + 2 \operatorname{Re} \sum_{k=1}^N a_k e^{j2\pi k f_0 t}$$

where $a_0 = A_0$ and $a_k = \frac{1}{2} A_k e^{j\phi_k}$ for $k \geq 0$

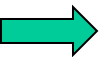
- **Synthesis of “ah” vowel from *SPFirst* (page 45)**

<http://dspfirst.gatech.edu/chapters/03spect/demos/vowel>

Matlab command: vowel

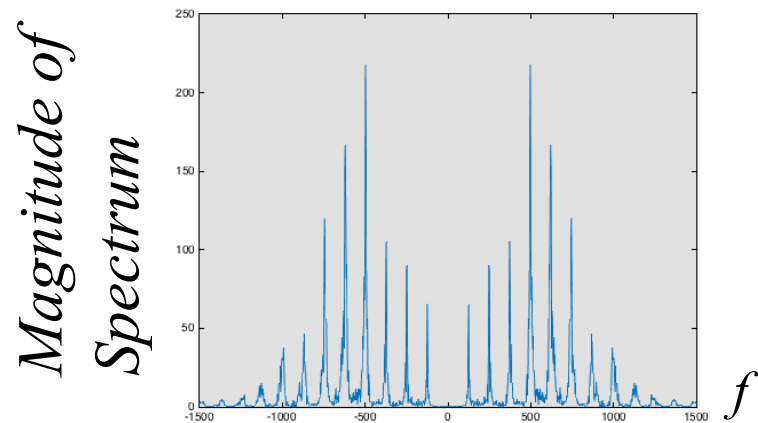
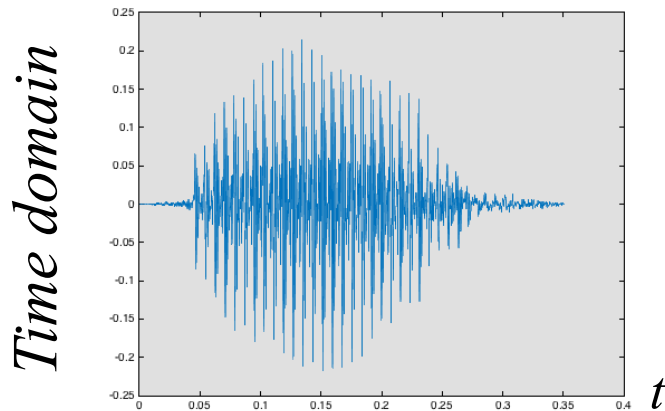
Comment out lines that start with *capture* and *imwrite*

Combines components at 200, 400, 500, 1600 and 1700 Hz



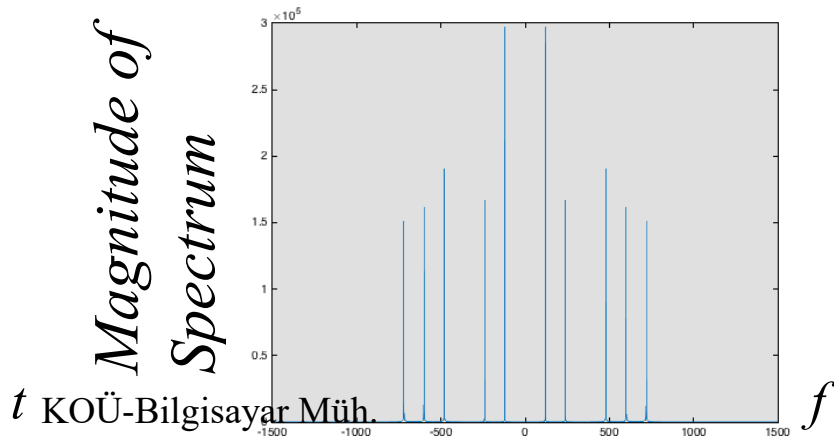
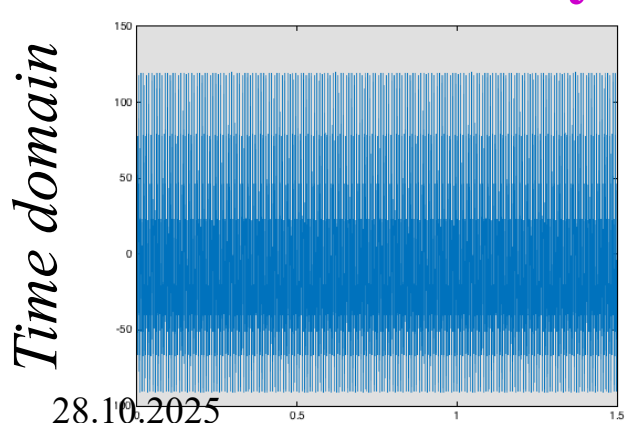
Speech Example Revisited

- A recording of the “ah” vowel being spoken



“ah”
recording

- How close did synthesized vowel sound like “ah”?



synthesized
sound

Fourier Series

- Any *periodic* signal $x(t)$ can be synthesized with a sum of harmonically related sinusoids, which is called as “Fourier Series Representation”



Joseph Fourier
lived from 1768 to 1830

- Mathematical theory realized by Fourier series:

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{j2\pi k f_0 t}$$

Complex Fourier Series Representation of $X(t)$,

$$C_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-j2\pi k f_0 t} dt$$

C_k is the Complex Fourier Series Coefficients

T_0 : fundamental period

f_0 : fundamental frequency ($f_0 = 1 / T_0$)

k -th complex exponential in summation has frequency $f_k = k f_0$

Compact Fourier Series Representation

In Complex F.S representation coefficients C_k are conjugate symmetric :

$$C_{-k} = C_k^* \text{ where } C_k = \frac{1}{2} c_k e^{j\phi_k}$$

So, ‘Complex F.S representation of $x(t)$ ’ can be converted into real valued ‘Compact Fourier Series representation’

$$x(t) = c_0 + \sum_{k=1}^{\infty} c_k \cos(2\pi k f_0 t + \theta_k)$$

General Fourier Series Representation

$$x(t) = a_0 + \sum_{k=1}^{\infty} a_n \cos(k\omega_0 t) + b_n \sin(k\omega_0 t)$$

$$a_0 = \frac{1}{T_0} \int_0^{T_0} f(t) dt$$

$$a_k = \frac{2}{T_0} \int_0^{T_0} f(t) \cos(k\omega_0 t) dt$$

$$b_k = \frac{2}{T_0} \int_0^{T_0} f(t) \sin(k\omega_0 t) dt$$

The coefficients are related to Compact F.S. representation

$$x(t) = c_0 + \sum_{k=1}^{\infty} c_k \cos(n\omega_0 t + \theta_k)$$

where $c_0 = a_0$, $c_k = \sqrt{a_k^2 + b_k^2}$, and

$$\theta_n = \tan^{-1} \left(\frac{b_k}{a_k} \right)$$

Relations

- **Fundamental** $x_1(t) = c_1 \cos(2 \pi f_0 t + \theta_1)$

Fundamental frequency in Hertz is f_0

Fundamental frequency in rad/s is $\omega_0 = 2 \pi f_0$

- **Harmonic** $x_k(t) = c_k \cos(2 \pi k f_0 t + \theta_k)$

Frequency, $k f_0$, is k -th harmonic of f_0

- **Magnitude/phase and Cartesian representations**

$$\begin{aligned} c_k \cos(k \omega_0 t + \theta_k) &= \\ c_k \cos(\theta_k) \cos(k \omega_0 t) - c_k \sin(\theta_k) \sin(k \omega_0 t) &= \\ a_k \cos(k \omega_0 t) &+ b_k \sin(k \omega_0 t) \end{aligned}$$

Existence of the Fourier Series

- **Existence**

$$\int_0^{T_0} |x(t)| dt < \infty$$

- **Convergence for all t**

$$|x(t)| < \infty \quad \forall t$$

- **Finite number of maxima and minima in one period of $x(t)$**

- **What about periodic extensions of**

$$g(t) = \frac{1}{t} \quad \text{for } -1 \leq t < 1 \quad \text{KOÜ-Bilgisayar Müh.} \quad s(t) = \sin\left(\frac{1}{t}\right) \quad \text{for } 0 \leq t < 1$$

How to Calculate Fourier Series?

- **Analysis: start with $x(t)$ and compute $\{ C_k \}$**

Integrate $x(t)$ over fundamental period T_0

$$C_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-j2\pi k f_0 t} dt$$

Calculation of c_0 simplifies to average value of $x(t)$

$$C_0 = \frac{1}{T_0} \int_0^{T_0} x(t) dt$$

Can often “eyeball” C_0 without performing integration

Example #1: With $x(t) = \cos(2\pi f_0 t)$, what is C_0 ?

Example #2: With $x(t) = \cos^2(2\pi f_1 t)$, what is C_0 ?

- **Synthesis: start with $\{ C_k \}$ and compute $x(t)$**

General case:
$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{j2\pi k f_0 t}$$

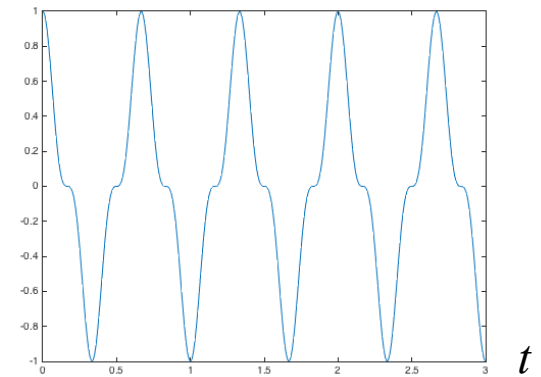
Example #1

- Find Fourier series coefficients for $x(t) = \cos^3(3\pi t)$

Approach #1: Fourier analysis formulas

Plot $x(t)$ to find $T_0 = 3 \text{ s} / 4.5 = 2/3 \text{ s}$
and “eyeball” $C_0 = 0$ and then

$$C_k = \frac{1}{T_0} \int_0^{T_0} \cos^3(3\pi t) e^{-j2\pi k f_0 t} dt$$



Approach #2: Expand into complex exponentials

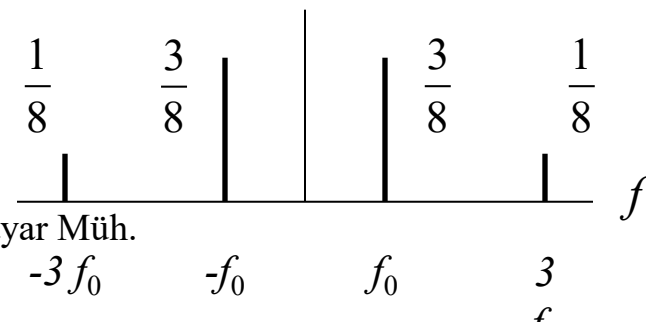
$$x(t) = \frac{e^{j3\pi t} + e^{-j3\pi t}}{2} = \frac{1}{8} (e^{j9\pi t} + 3e^{j3\pi t} + 3e^{-j3\pi t} + e^{-j9\pi t})$$

- Resulting spectrum

$$\omega_0 = \gcd(3\pi, 9\pi) = 3\pi$$

$$f_0 = 1.5 \text{ Hz}$$

Spectrum

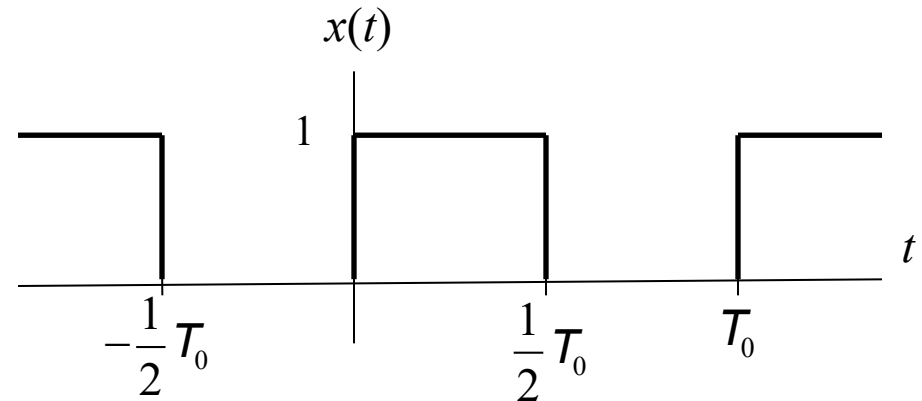


Example #2: Fourier Analysis of a Square Wave

- Periodic square wave with 50% duty cycle**

Defined for one period as

$$x(t) = \begin{cases} 1 & \text{for } 0 \leq t < \frac{1}{2} T_0 \\ 0 & \text{for } \frac{1}{2} T_0 \leq t < T_0 \end{cases}$$



- Fourier coefficients**

1. $C_0 = 1/2$ because $x(t)$ is 1 half the time and 0 half the time

2. Then,
$$C_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-j2\pi k f_0 t} dt = \frac{1}{T_0} \int_0^{\frac{1}{2} T_0} e^{-j2\pi k f_0 t} dt = \left(\frac{1}{T_0} \right) \frac{e^{-j2\pi k f_0 t}}{-j2\pi k f_0} \Bigg|_0^{\frac{1}{2} T_0}$$

$$C_k = \left(\frac{1}{T_0} \right) \frac{e^{-j2\pi k f_0 (T_0/2)} - e^{-j2\pi k f_0 (0)}}{-j2\pi k f_0} = -\frac{e^{-j\pi k} - 1}{j2\pi k} = \frac{1 - (-1)^k}{j2\pi k}$$

For $k \neq 0$

Spectrum for a Square Wave

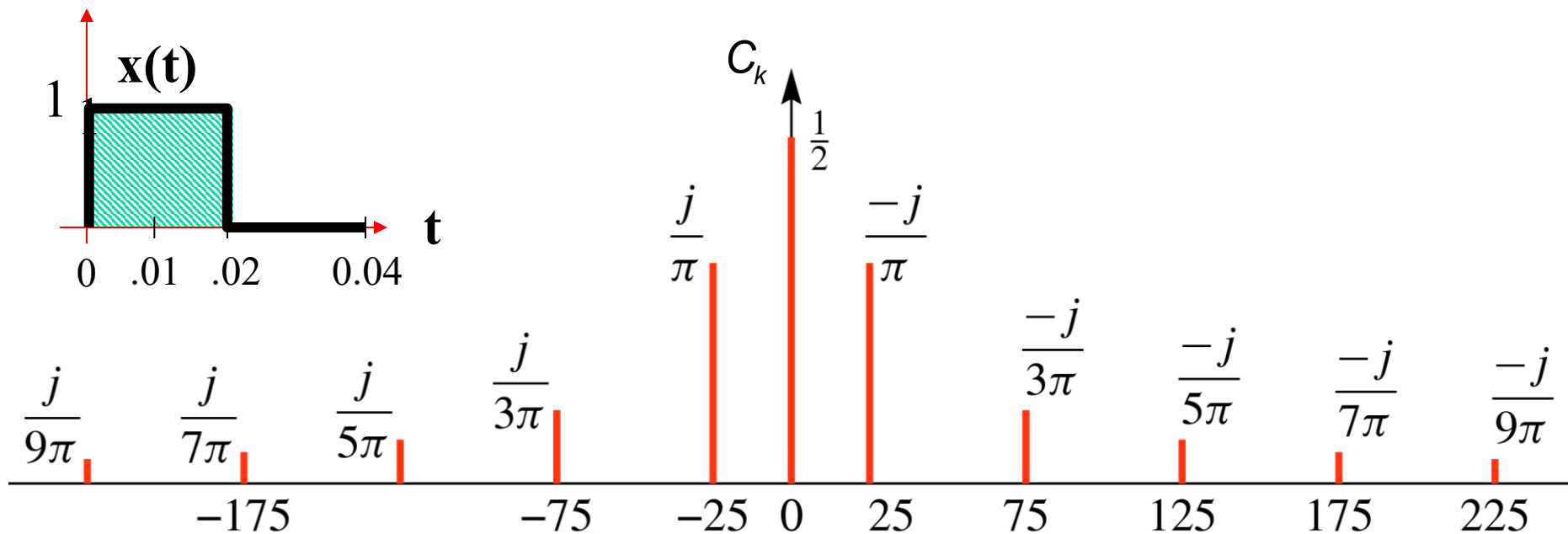
- Fourier coefficients**

Independent of T_0

- Example**

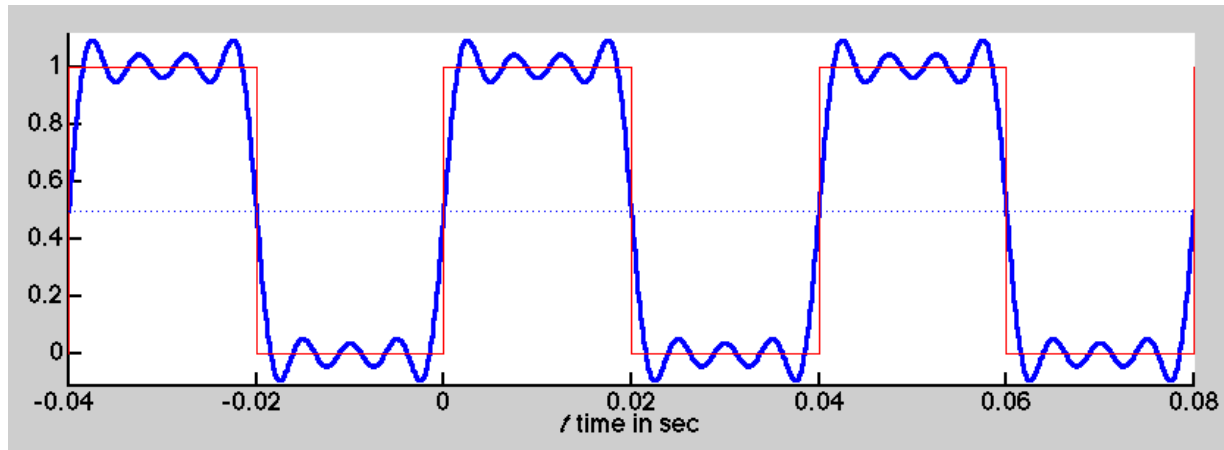
$$T_0 = 0.04 \text{ s} \rightarrow f_0 = 25 \text{ Hz}$$

$$C_k = \begin{cases} \frac{1}{2} & \text{for } k=0 \\ 0 & \text{for } k \text{ even but not zero} \\ -\frac{j}{\pi k} & \text{for } k \text{ odd} \end{cases}$$

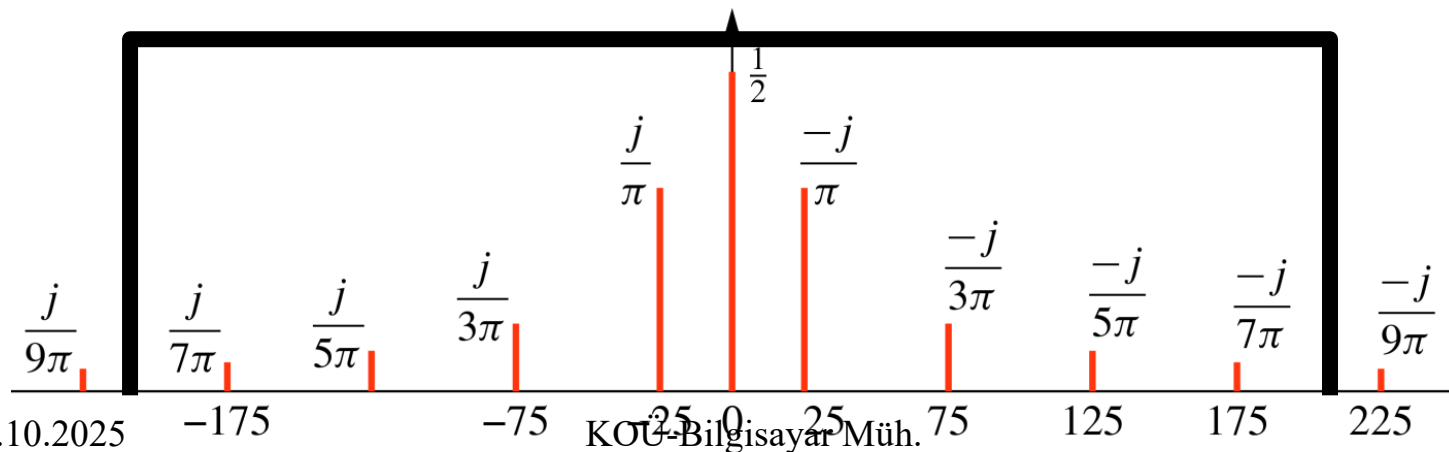


Fourier Synthesis of a Square Wave

- Synthesis using up to the 7th harmonic



$$y(t) = \frac{1}{2} + \frac{2}{\pi} \sin(50\pi t) + \frac{2}{3\pi} \sin(150\pi t) + \frac{2}{5\pi} \sin(250\pi t) + \frac{2}{7\pi} \sin(350\pi t)$$



demo

Fourier Synthesis vs. Analysis

- **Fourier Synthesis**

Given (f_k, c_k, ϕ_k) values, create $x(t)$

$$x(t) = c_0 + \sum_{k=1}^N c_k \cos(2\pi f_k t + \phi_k)$$

Implementing synthesis formula is somewhat *straightforward*

Achieving high perceptual quality in $x(t)$ is *difficult*, e.g. synthesized speech or music

- **Fourier Analysis**

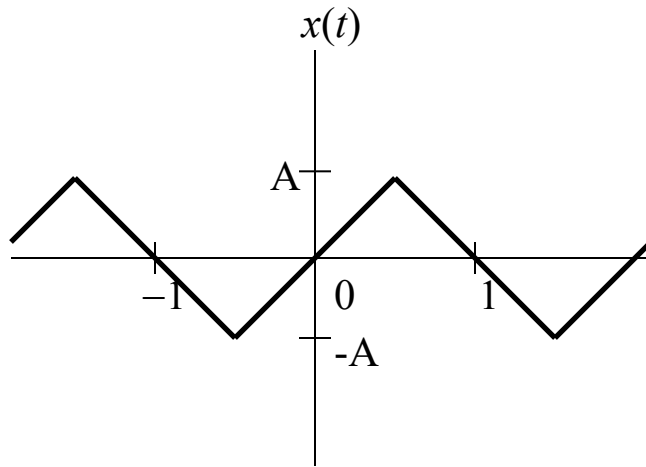
Very *difficult* task, esp. for physical signals

Given $x(t)$, extract (f_k, c_k, ϕ_k) values. How many?

Using (f_k, c_k, ϕ_k) , how close is model to signal?

Need mathematical algorithms for computer

Example #3



- Fundamental period**

$$T_0 = 2$$

- Fundamental frequency**

$$f_0 = 1/T_0 = 1/2 \text{ Hz}$$

$$\omega_0 = 2\pi/T_0 = \pi \text{ rad/s}$$

28.10.2025

$$x(t) = a_0 + \sum_{k=1}^{\infty} a_k \cos(\pi k t) + b_k \sin(\pi k t)$$

$$a_0 = 0 \quad (\text{by inspection of the plot})$$

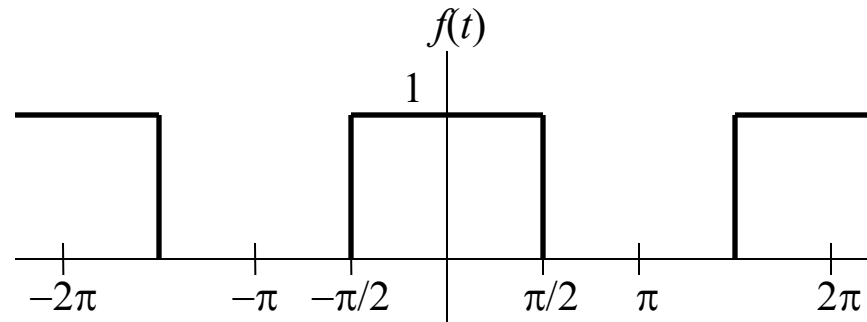
$$a_k = 0 \quad (\text{because it is odd symmetric})$$

$$b_k = \int_{-1/2}^{1/2} 2A t \sin(\pi k t) dt +$$

$$\int_{3/2}^{5/2} (2A - 2A t) \sin(\pi k t) dt$$

$$b_k = \begin{cases} 0 & k \text{ is even} \\ \frac{8A}{n^2 \pi^2} & n = 1, 5, 9, 13, \dots \\ -\frac{8A}{n^2 \pi^2} & n = 3, 7, 11, 15, \dots \end{cases}$$

Example #4



- Fundamental period**

$$T_0 = 2\pi$$

- Fundamental frequency**

$$f_0 = 1/T_0 = 1/(2\pi) \text{ Hz}$$

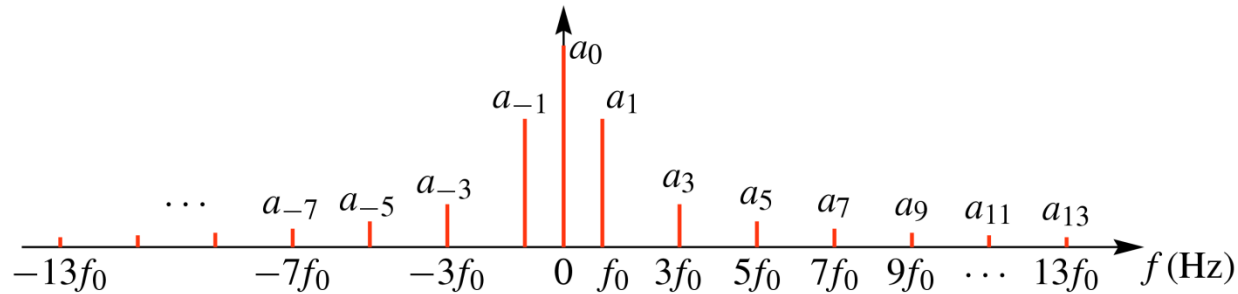
$$\omega_0 = 2\pi/T_0 = 1 \text{ rad/s}$$

$$c_0 = \frac{1}{2}$$

$$c_k = \begin{cases} 0 & k \text{ even} \\ \frac{2}{\pi k} & k \text{ odd} \end{cases}$$

$$\theta_k = \begin{cases} 0 & \text{for all } k \neq 3, 7, 11, 15, \dots \\ -\pi & k = 3, 7, 11, 15, \dots \end{cases}$$

Spectrum & Fourier Series



Spectrum Plot
 (a_k, kf_0) versus f

$T_0 = \text{Period}$

$N = \text{Number of Coefficients}$

